Interview questions and answers

Q1. What is EDA and why is it important?

EDA stands for Exploratory Data Analysis. It is the process of analysing and understanding a dataset using statistical methods and visualizations.

It is important because:

- Helps identify missing values.
- Detects outliers.
- Finds patterns and trends.
- Understands relationships between variables.
- Helps prepare data for machine learning.

Q2. Which plots do you use to check correlation?

I use the following visualizations to understand relationship and correlations between variables:

- Heatmap – to visualize correlation between multiple numerical variables.
- Scatterplot – to identify relationships between two variables.
- Pairplot – to visualize relationships between multiple numerical variables.

Q3. How do you handle skewed data?

Skewed data can be handled using:

- Log transformation.
- Square root transformation.
- Box-Cox transformation.
- Removing or treating extreme outliers when justified.

For example, np.log1p() can reduce right skewness.

Q4. How do you detect multicollinearity?

Multicollinearity occurs when two or more independent variables are highly correlated.

I detect multicollinearity using:

1. Correlation matrix.

2. Heatmap.

3. Variance Inflation Factor (VIF).

A high VIF value indicates possible multicollinearity. Generally, VIF above 5 deserves investigation, while above 10 is often considered serious.

To handle it, I may remove one highly correlated variable or use dimensionality reduction techniques.

Q5.What are univariate, bivariate, and multivariate analyses?

These are three important types of data analysis.

Analysis	Meaning	Example
Univariate	Analysis of one variable	Distribution of passenger age
Bivariate	Analysis of two variables	Gender vs survival
Multivariate	Analysis of three or more variables	Gender, passenger class, and survival

Univariate analysis helps understand individual variables, bivariate analysis identifies relationships, and multivariate analysis helps understand complex patterns.

Q6. Difference between heatmap and pairplot?

A heatmap and pairplot are both visualization techniques, but they serve different purposes.

Heatmap	Pairplot
Uses colors to represent values	Uses multiple graphs
Commonly shows correlation	Shows pairwise distributions and relationships
Easy to identify strong correlations	Useful for identifying patterns and clusters
Compact visualization	Can become crowded with many variables

Q7. How do you summarize your insights?

I summarize insights by examining statistical results and visualizations.

My summary includes:

- Important patterns and trends.
- Relationships between variables.
- Missing values and outliers.
- Distribution of numerical and categorical variables.
- Business or real-world implications.

Example: In the Titanic dataset, female passengers had a higher survival rate than male passengers. First-class passengers also had a higher survival rate compared with passengers in lower classes.

Finally, I present the findings using charts, statistical summaries, and clear conclusions.